

Thanuish Kumar Sathiaraj

AI & LLM ENGINEER | AGENTIC SYSTEMS | COMPUTER VISION

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M.Sc. Computer Science student specializing in computer vision and moving toward Physical AI: perception systems that understand the real world, combined with large language models. At the Max Planck Institute I build vision pipelines and the backend software behind them (Python, Flask, REST APIs, Docker). Alongside that I build many projects on my own, purely to learn and understand things properly, from agentic AI applications using LangGraph and the Model Context Protocol (MCP), including building my own MCP servers, to LLM routing and evaluation systems.

PROFESSIONAL EXPERIENCE

Student Research Assistant, AI & Software Engineering

Max Planck Institute for Intelligent Systems (Perceiving Systems Dept.), Tübingen, Germany | Nov 2025 – Apr 2026

- **Backend development:** developed backend services and internal web tooling with Python and the Flask framework, exposing research pipelines through REST APIs so newly uploaded data is ingested, processed and returned automatically, covering the full path from upload to model output.
- **Model serving:** integrated machine learning models into a live internal service so new data is processed and reflected automatically, taking models out of notebooks and into something a team actually uses.
- **Data pipelines & database:** built and maintained ETL pipelines that collected, cleaned and validated large datasets into structured, queryable form, with database development and output visualization for downstream use.
- **Deployment & infrastructure:** ran the system as containerized Docker services on Linux, staged across components so fault isolates to a single stage instead of taking the pipeline down.
- **Evaluation & QA:** reviewed model outputs, separated successful cases from failures, diagnosed root causes, and fed concrete improvement proposals back to researchers for the next iteration.
- **Computer vision:** built computer vision pipelines and a real-time viewer for 3D human-body reconstruction from multi-camera video using SMPL-X parametric body models, supporting PhD research on MAMMA (Markerless Accurate Multi-Person Motion Acquisition).
- **Engineering process:** worked in an Agile/Scrum team, tracked work with Jira, used Git and GitHub with branching, pull requests and code review, and maintained shared technical documentation. Referred by Tsvetelina Alexiadis, Data Team Lead.

Machine Learning Engineer Intern

Shiash Info-Solutions | Apr 2024 – Feb 2025

- Built an end-to-end CNN Pipeline system in Python, owning the full lifecycle from data collection and cleaning through model training, validation and deployment.
- Improved the model with Bayesian hyperparameter optimization, data augmentation, normalization and dropout regularization, and built automated preprocessing pipelines that cut training time by roughly 20%.
- **Stack:** Python, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas.

PROJECTS

AI Autopilot: Semantic Caching & Intelligent LLM Routing Gateway (in development, 2026)

Python, FastAPI, Pydantic v2, Valkey, Sentence-Transformers, Ollama, PostgreSQL, scikit-learn, Streamlit, Docker Compose, pytest

- Building a production-style AI gateway (mainly for Ecommerce) that answers each request by the cheapest sufficient path: semantic cache first (embedding similarity search in Valkey), then a deterministic knowledge base, and only then an LLM, so unnecessary inference is avoided entirely rather than merely made cheaper.
- Routing layer classifies request complexity with a scikit-learn model and selects a local Ollama model accordingly, behind a provider abstraction so cloud providers can be added without touching the routing logic.
- Asynchronous quality verification scores each answer and automatically escalates failures to a stronger model; every request is logged to PostgreSQL and surfaced in a Streamlit dashboard, benchmarked against an all-strong-model baseline.

Market Sentinel: Multi-Agent AI System with MCP and RAG

Python, LangGraph, MCP, RAG, Pinecone, pytest, Git | github.com/Thanuish/Market_sentinal

- Built a multi-agent AI system in LangGraph where agents coordinate through explicit orchestration and the final action is structurally unreachable until an evaluator agent writes approval into graph state, so no prompt can talk the system past its own gate.
- Exposed the deterministic decision logic as a Model Context Protocol (MCP) server tool with API-style calls, and added retrieval-augmented generation (RAG) backed by a Pinecone vector database, so the model works from retrieved evidence it cannot silently alter.
- Proved less LLM Hallucination
- Backed the pipeline with a 12-test pytest suite proving an unapproved input cannot reach execution.

Sentiment Transfer with LLMs: Fine-Tuning vs Prompting Benchmark

Python, Hugging Face, T5/FLAN-T5, LLaMA 3, Ollama, BLEU, Sentence-Transformers

- Benchmarked a statistical baseline against fine-tuned T5/FLAN-T5 and prompted LLaMA 3 served locally through Ollama, implementing the methods from research papers.
- Went beyond training into prompt engineering: iterated on prompts, in-context examples and instructions to steer model behaviour, and measured whether a well-prompted model could beat a fine-tuned one.
- Scored outputs on a 147-pair dataset with BLEU, embedding similarity and a 50+ annotator human study; the prompted LLM outperformed fine-tuning under low-resource conditions, a direct cost-versus-quality result.

Deepfake Detection in Images and Video (Bachelor's Thesis)

Python, TensorFlow, Keras, OpenCV, MTCNN, Inception-ResNet, LSTM, Grad-CAM, SHAP

- Built, as sole developer, a cascading vision pipeline (MTCNN face detection, Inception-ResNet CNN features, classification, LSTM for temporal information across video frames) reaching 91.7% accuracy and F1 0.89, a 12.4% improvement over standalone models, over 140GB+ of image and video data on Linux, with Grad-CAM and SHAP interpretability.

SKILLS

LLMs & Agentic AI: LangGraph multi-agent orchestration, MCP (Model Context Protocol) server design and development, RAG with Pinecone vector database, semantic caching and embedding similarity search, LLM routing and evaluation, prompt engineering, LLM APIs (OpenAI/ChatGPT, Anthropic), local model serving with Ollama, fine-tuning T5/FLAN-T5 and LLaMA 3 (Hugging Face), Sentence-Transformers

Backend & Software Engineering: Python (primarily for Intergration), SQL, C, C#, JavaScript, Bash; Flask and FastAPI, REST API design, Pydantic schema validation, React front-end, HTML/CSS, ETL and data pipelines, database development (PostgreSQL, SQLite), Valkey/Redis

Infrastructure & Practice: Docker and Docker Compose on Linux, staged deployment for fault isolation, Git/GitHub with pull requests and code review, Jira, Agile/Scrum, pytest unit and integration testing, technical documentation; building CI/CD pipelines

ML & Computer Vision: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, OpenCV; CNN, LSTM, Inception-ResNet, object detection (YOLOv6), hyperparameter optimization, data augmentation, model evaluation, interpretability (Grad-CAM, SHAP), Streamlit and Plotly dashboards

EDUCATION

M.Sc. Computer Science, University of Stuttgart, Germany | Apr 2025 – Apr 2028 (expected) | Coursework: Reinforcement Learning, Knowledge Graphs, Computer Vision

B.E. Computer Science & Engineering, Anna University | Aug 2020 – Jun 2024 | Coursework: Machine Learning & AI, DBMS, Operating Systems | Languages: English C1, German A2 (studying)